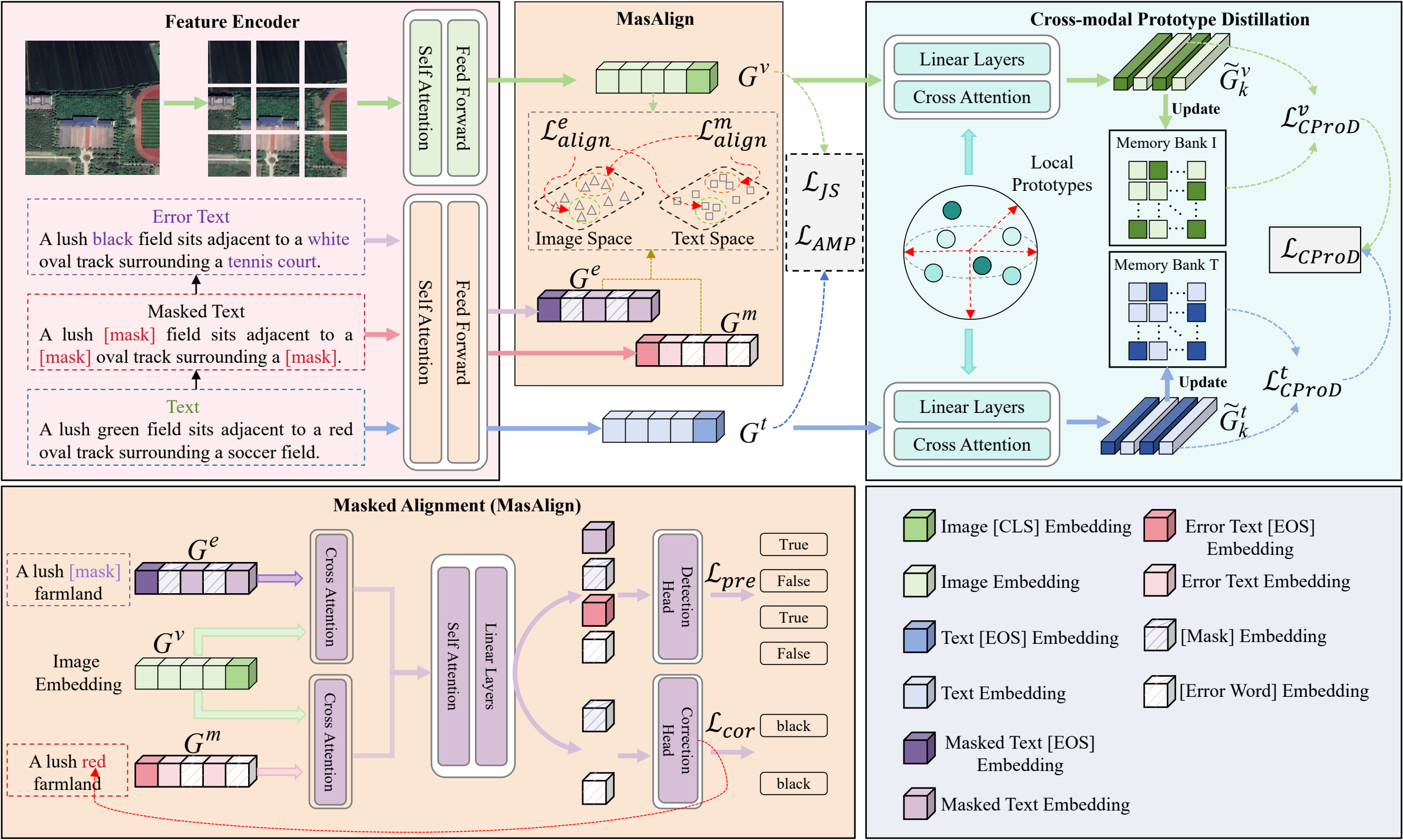

Feature Encoder
Self Attention
Feed Forward
Error Text
A lush black field sits adjacent to a white oval track surrounding a tennis court.
Masked Text
A lush [mask] field sits adjacent to a [mask] oval track surrounding a [mask].
Text
A lush green field sits adjacent to a red oval track surrounding a soccer field.
MasAlign
G^v
$\mathcal{L}^e_{align}$
$\mathcal{L}^m_{align}$
Image Space
Text Space
G^e
G^m
G^t
$\mathcal{L}_{JS}$
$\mathcal{L}_{AMP}$
Cross-modal Prototype Distillation
Linear Layers
Cross Attention
$\widetilde{G}^v_k$
Update
Local Prototypes
Memory Bank I
Memory Bank T
$\mathcal{L}^v_{CProD}$
$\mathcal{L}_{CProD}$
$\mathcal{L}^t_{CProD}$
Masked Alignment (MasAlign)
A lush [mask] farmland
Image Embedding
A lush red farmland
Cross Attention
Self Attention
Linear Layers
Detection Head
$\mathcal{L}_{pre}$
True
False
Correction Head
$\mathcal{L}_{cor}$
black
Image [CLS] Embedding
Error Text [EOS] Embedding
Image Embedding
Error Text Embedding
Text [EOS] Embedding
[Mask] Embedding
Text Embedding
[Error Word] Embedding
Masked Text [EOS] Embedding
Masked Text Embedding